Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 8: Reading a Formula off a Table

Disjunctive Normal Form

Lehman, Leighton & Meyer, *MCS* Section 3.4.1

Name: _____ Date: _____

A change of textbook, for two days only

Chartrand does not cover normal forms, so Wednesday and Friday use the free supplement. Everything else this semester stays in Chartrand.
One vocabulary note. Where Chartrand says a compound statement is a *tautology*, the supplement says the formula is **valid**. Same idea, two names. It also says **satisfiable** for a formula that is true in *at least one* row.

The question for today

So far you have always been handed a formula and asked to build its table. Today we run that backwards.

> Given *any* column of T's and F's, can you write down a formula that produces it?

The answer is yes, always, and there is a recipe.

By the end of today you should be able to

1. Say what a literal, an AND-term, and a disjunctive normal form are.
2. Read a DNF directly off a truth table.
3. Explain why every compound statement has one.

1. Vocabulary

Three words

A **literal** is a variable or its negation: P or $\sim P$.
An **AND-term** is a conjunction of literals: $P \wedge (\sim Q) \wedge R$.
A **disjunctive normal form (DNF)** is a disjunction of AND-terms, where each AND-term mentions *every* variable exactly once — either plain or negated.

So a DNF looks like an OR of ANDs. Some people call it *sum of products*, treating $\vee$ as $+$ and $\wedge$ as $\times$ — the same analogy from Lecture 3.

Which of these are in DNF over $\{P, Q\}$?

Formula	DNF?	Why
$(P \wedge Q) \vee (\sim P \wedge Q)$	Yes	OR of AND-terms; each names both variables.
$P \vee (\sim P \wedge Q)$	No	First term omits Q.
$\sim(P \wedge Q)$	No	Negation applied to a compound, not to a variable.

2. The recipe

Reading a DNF off a truth table

1. Look at each row where the formula is **true**.
2. For that row, write an AND-term: each variable *plain* if it is T in that row, *negated* if it is F.
3. OR all those AND-terms together.

Why it works. Each AND-term is true in exactly one row — the row it came from. So the OR of them is true in exactly the rows you selected, and false everywhere else. That is the column you wanted.

3. Worked together: $P \Rightarrow Q$

P	Q	$P \Rightarrow Q$	AND-term for the true rows
T	T	T	$P \wedge Q$
T	F	F	—
F	T	T	$(\sim P) \wedge Q$
F	F	T	$(\sim P) \wedge (\sim Q)$

$$P \Rightarrow Q \equiv (P \wedge Q) \vee ((\sim P) \wedge Q) \vee ((\sim P) \wedge (\sim Q))$$

Three true rows, so three AND-terms.

The number of terms is not a mystery

The DNF has exactly one AND-term per **true** row. Count the T's in the column and you know how many terms to expect before you write anything.

4. Your turn: exclusive or

P	Q	$P \oplus Q$	AND-term
T	T	F	—
T	F	T	$P \wedge (\sim Q)$
F	T	T	$(\sim P) \wedge Q$
F	F	F	—

$$P \oplus Q \equiv (P \wedge (\sim Q)) \vee ((\sim P) \wedge Q)$$

You have written this before

That is the answer to Practice Set 3, Problem 11 — "either P without Q, or Q without P." Back then it took a truth table to check. Now the recipe produces it directly.

5. Three variables

Build the DNF for $A \wedge (B \vee C)$.

A	B	C	$B \vee C$	$A \wedge (B \vee C)$	AND-term
T	T	T	T	T	$A \wedge B \wedge C$
T	T	F	T	T	$A \wedge B \wedge (\sim C)$
T	F	T	T	T	$A \wedge (\sim B) \wedge C$
T	F	F	F	F	—
F	T	T	T	F	—
F	T	F	T	F	—
F	F	T	T	F	—
F	F	F	F	F	—

Three true rows again, so the DNF has three AND-terms, each naming all three variables.

DNF is not the same as short

The original formula $A \wedge (B \vee C)$ is far shorter than its DNF. Normal form is about being in a *standard shape*, not about being small.
The payoff is that two formulas are equivalent **if and only if** they have the same DNF (once the terms are sorted). That gives you a way to test equivalence that does not depend on noticing a clever rewrite.

6. The theorem

MCS Theorem 3.4.3

Every compound statement is logically equivalent to a disjunctive normal form.

The proof is the recipe: any column of T's and F's can be reproduced by ORing one AND-term per true row.

One edge case. What is the DNF of a *contradiction*? There are no true rows, so there are no terms. We write it as **F**.

Exit ticket

1. How many AND-terms will the DNF of $P \Leftrightarrow Q$ have? Explain without writing it out.

2. Write the DNF of $\sim(P \vee Q)$.

Before Friday

Read *MCS* Section 3.4.1 again through the conjunctive form paragraph (pp. 51–52), and Section 3.3.2, "Validity and Satisfiability" (p. 50).

Self-check (not collected). For each of $P \wedge Q$, $\sim P$, and $(\sim P) \vee Q$, write the DNF over the variables P and Q. Check each against a truth table.

Homework 3 is due Friday, September 4, 11:59 p.m.

Sections 1.3 and 1.4. One PDF to Canvas with the collaboration and AI statement first.

Questions I still have